Lemma: $R$是交换环，$I$是$R$的唯一的素理想，则有：$I$是$R$的极大理想

Proof: 假设$R$是零环，则$R=\{0_R\}$ $\therefore$ $R$的理想只有$R$.

$\therefore$ $R$没有真理想 $\therefore$ $R$没有素理想 矛盾. $\therefore$ $R$是非零环

$\because$ $R$是非零交换环 $\therefore$ $R$至少有1个极大理想

设$M$是$R$的极大理想.

$\because$ $R$是交换环，$M$是$R$的极大理想 $\therefore$ $M$是$R$的素理想

$\because$ $I$是$R$的唯一的素理想 $\therefore$ $M=I$

$\therefore$ $I$是$R$的极大理想. $\square$

Lemma: $R$是交换环，$I$是$R$的唯一的素理想，则有：$R/I$是域.

Proof: $\because$ $R$是交换环，$I$是$R$的唯一的素理想

$\therefore$ $I$是$R$的极大理想

$\therefore$ $R/I$是域. $\square$

Lemma: $R$是环，对$\forall x\in R$，若$x$是$R$的幂零元，则有：$1_R-x\in R^\times$且$1_R+x\in R^\times$

Proof: 对$\forall x\in R$，若$x$是$R$的幂零元，则有：

$\because x$是$R$的幂零元 $\therefore \exists n\in\mathbb{Z}_{\geq 1}$, s.t. $x^n=0_R$. $\because n\in\mathbb{Z}_{\geq 1}$ $\therefore n-1\in\mathbb{Z}_{\geq 0}$

$\therefore x^{n-1}\in R$ $\therefore 1_R+x+\cdots+x^{n-1}\in R$

$\because (1_R-x)\cdot(1_R+x+\cdots+x^{n-1})=1_R\cdot(1_R+x+\cdots+x^{n-1})-x(1_R+x+\cdots+x^{n-1})$

$=1_R+x+\cdots+x^{n-1}-x-x^2-\cdots-x^{n-1}-x^n=1_R-x^n=1_R-0_R=1_R$

$(1_R+x+\cdots+x^{n-1})(1_R-x)=(1_R+x+\cdots+x^{n-1})\cdot 1_R-(1_R+x+\cdots+x^{n-1})x$

$=1_R+x+\cdots+x^{n-1}-x-x^2-\cdots-x^{n-1}-x^n=1_R-x^n=1_R-0_R=1_R$

$\therefore 1_R-x\in R^\times$ 且 $(1_R-x)^{-1}=1_R+x+\cdots+x^{n-1}$

$\because 1_R-x+\cdots+(-1)^{n-1}x^{n-1}\in R$，且有：

$(1_R+x)(1_R-x+\cdots+(-1)^{n-1}x^{n-1})=1_R-x+\cdots+(-1)^{n-1}x^{n-1}+x-x^2+\cdots$

$+(-1)^{n-2}x^{n-1}+(-1)^{n-1}x^n=1_R+(-1)^{n-1}x^n=1_R+(-1)^{n-1}\cdot 0_R=1_R$

$(1_R-x+\cdots+(-1)^{n-1}x^{n-1})(1_R+x)=1_R-x+\cdots+(-1)^{n-1}x^{n-1}+x-x^2+\cdots+(-1)^{n-2}x^{n-1}$

$+(-1)^{n-1}x^n=1_R+(-1)^{n-1}x^n=1_R+(-1)^{n-1}\cdot 0_R=1_R$

$\therefore 1_R+x\in R^\times$ 且 $(1_R+x)^{-1}=1_R-x+\cdots+(-1)^{n-1}x^{n-1}$ $\square$

Lemma: $R$是交换环，对$\forall x\in R$，若$x+nil(R)\in(R/nil(R))^\times$，则有：$x\in R^\times$.

Proof: $\because R$是交换环 $\therefore nil(R)$是$R$的理想. $\therefore R/nil(R)$是环

$\therefore R/nil(R)$是交换环.

对$\forall x\in R$, ~~若 $x\in$~~ $\because x\in R$ $\therefore x+nil(R)\in R/nil(R)$

若$x+nil(R)\in(R/nil(R))^{\times}$, 则存在$y+nil(R)\in R/nil(R)$ (其中$y\in R$), s.t. $(x+nil(R))(y+nil(R))=1_{R/nil(R)}=(y+nil(R))(x+nil(R))$.

$\therefore 1_R+nil(R)=1_{R/nil(R)}=(x+nil(R))(y+nil(R))=xy+nil(R)$

$\therefore 1_R\equiv_{nil(R)}xy$ $\therefore 1_R-xy\in nil(R)$ $\therefore 1_R-xy$是$R$的幂零元.

$\therefore 1_R-(1_R-xy)\in R^{\times}$

$\because 1_R-(1_R-xy)=1_R+(-(1_R-xy))=1_R+(-(1_R+(-xy)))$

$=1_R+((-1_R)+xy)=(1_R+(-1_R))+xy=0_R+xy=xy$

$\therefore xy\in R^{\times}$ $\therefore x\in R^{\times}$且$y\in R^{\times}$ $\therefore x\in R^{\times}$. $\square$

Lemma: $R$是非零交换环, 则有: 下列三个命题等价:

① $R/nil(R)$是域

② 对$\forall x\in R$, $x\in R^{\times}$或$x\in nil(R)$

③ $nil(R)$是$R$的唯一的素理想.

Proof: $\because R$是非零环 $\therefore 1_R\neq 0_R$ $\therefore$对$\forall n\in\mathbb{Z}_{\geq 1}$, 有: $1_R^n=1_R\neq 0_R$

$\therefore 1_R\notin nil(R)$ $\because R$是交换环 $\therefore nil(R)$是$R$的理想

$\therefore R/nil(R)$是交换环. $\therefore 1_R+nil(R)\neq 0_{R/nil(R)}$

$\therefore 1_{R/nil(R)}\neq 0_{R/nil(R)}$ $\therefore R/nil(R)$是非零环 $\therefore R/nil(R)$是非零交换环.

①⇒② 对$\forall x\in R$，有：$x+nil(R)\in R/nil(R)$

若$x+nil(R)=0_{R/nil(R)}$，则$x+nil(R)=0_R+nil(R)$

$\therefore x\equiv_{nil(R)}0_R$　　$\therefore x-0_R\in nil(R)$　　$\therefore x\in nil(R)$

若$x+nil(R)\neq 0_{R/nil(R)}$，则$\because R/nil(R)$是域

$\therefore x+nil(R)\in(R/nil(R))^{\times}$　　$\therefore x\in R^{\times}$.

$\therefore x\in R^{\times}$或$x\in nil(R)$.

②⇒③ $\because R$是交换环　　$\therefore nil(R)$是$R$的理想.

$\because 1_R\in R$且$1_R\notin nil(R)$　　$\therefore nil(R)\subsetneq R$　　$\therefore nil(R)$是$R$的真理想

对$\forall\lambda,\mu\in R$，若$\lambda\mu\in nil(R)$，则有：

若$\lambda\in nil(R)$，则$\lambda\in nil(R)$

若$\lambda\notin nil(R)$，则$\because\lambda\in R$且$\lambda\notin nil(R)$　　$\therefore\lambda\in R^{\times}$　　$\therefore\lambda^{-1}\in R$

$\because nil(R)$是$R$的理想，$\lambda^{-1}\in R$　　$\therefore\lambda^{-1}nil(R)\subseteq nil(R)$

$\therefore\lambda^{-1}(\lambda\mu)\in\lambda^{-1}nil(R)\subseteq nil(R)$

$\because\lambda^{-1}(\lambda\mu)=(\lambda^{-1}\lambda)\mu=1_R\cdot\mu=\mu$　　$\therefore\mu\in nil(R)$

$\therefore\lambda\in nil(R)$或$\mu\in nil(R)$

$\therefore nil(R)$是$R$的素理想.

设$B$是$R$的一个任意的素理想.

$\because R$是非零交换环　　$\therefore nil(R)=\bigcap_{p\in Spec(R)}p$

$\because B\in Spec(R)$　　$\therefore nil(R)\subseteq B$.

对$\forall x\in B$，假设$x\notin nil(R)$，则有：$\because x\in B$，$B$是$R$的素理想

$\therefore x \in R$. $\because x \in R$ 且 $x \notin nil(R)$ $\therefore x \in R^{\times}$ $\therefore x \in B \cap R^{\times}$

$\therefore B \cap R^{\times} \neq \emptyset$ $\therefore B = R$. $\because B$ 是 $R$ 的素理想 $\therefore B \neq R$. 矛盾.

$\therefore x \in nil(R)$ $\therefore B \subseteq nil(R)$

$\therefore B = nil(R)$

$\therefore nil(R)$ 是 $R$ 的唯一的素理想.

③ ⇒ ①: $\because R$ 是交换环, $nil(R)$ 是 $R$ 的唯一的素理想

$\therefore R/nil(R)$ 是域. $\square$